

THE ABSORPTION OF SOUND BY HELMHOLTZ RESONATORS WITH AND WITHOUT FLOW

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Abstract

Helmholtz resonators are commonly used as absorbers of incident acoustic power. Theoretical and experimental investigations have been performed in the four cases of no mean flow, grazing mean flow, bias mean flow and a combination of grazing and bias mean flows. In the absence of a mean flow, the absorption coefficient (defined as the proportion of incident energy absorbed) is a non-linear function of the acoustic pressure and high incident acoustic pressures are required before the absorption becomes significant. In contrast, when there is a mean flow present, either grazing or bias, the absorption is linear and thus absorption coefficient is independent of the magnitude of the acoustic pressure, and absorption is obtained over a wider range of frequencies. Non-linear effects are only discernible very close to resonance and at very high amplitude. With grazing mean flow, there is the undesirable effect that sound can be generated over a range of frequencies due to the interaction between the unsteadily shed vorticity waves and the downstream edge of the aperture. This production is not observed when there is a bias flow because here the vorticity is shed all around the rim of the aperture and swept away by the mean flow. When there is both a grazing mean flow and a mean bias flow, we find that only a small amount of bias mean flow, compared with grazing mean flow, is required to destroy the production of acoustic energy.

Introduction

Concern for the environment has resulted in increased legislative pressure on industrial gas turbine manufacturers to reduce pollutant emissions. This in turn has led to the adoption of lean premixed combustion systems. Lean premixed combustion offers many advantages in this respect such as lower soot and CO. Furthermore, combustion takes place at lower temperatures than conventional combustion systems, resulting in a dramatic reduction in thermal NOx emission. Unfortunately lean premixed combustion also suffers from some serious difficulties resulting from increased susceptibility to external disturbances

and acoustic perturbation and reduced flame stability. Moderate coupling between acoustic fluctuations and heat release can lead to unacceptable noise levels, whilst strong coupling can lead to self excited oscillations, the amplitude of which is so great that structural damage can occur.

There is, therefore, a clear need for practical methods of controlling the oscillations. In this project, the first six months of which are reported here, the performance of acoustic absorbers both with and without flow is investigated.

Helmholtz Resonators

Introduction

A commonly used device to absorb acoustic energy is a Helmholtz resonator. Resonance is established when the mass of the plug of air in the neck oscillates against the large volume of air in the chamber. At resonance small pressure perturbations give rise to large oscillating mass fluxes in the neck. These in turn induce large viscous losses and so narrow band sound absorption is achieved for frequencies close to resonance. Classical analysis¹⁰ uses a lumped element approach in which the specific geometry of the resonator is ignored and theory developed based upon the lumped effects of a neck of finite length and area and a cavity of fixed volume, giving a resonant frequency of $\omega = \bar{c}\sqrt{\frac{S}{Vl}}$, where S and l are the area and effective length of the neck (including end effects) respectively and V is the volume of the chamber.

The large acoustic absorption associated with Helmholtz resonators makes them suitable for the control of shock waves,³⁸ boundary layer turbulence^{15,16} and acoustic noise.²³ In this paper our main objective is to study the acoustic absorption of Helmholtz resonators, using a lumped parameter approach to quantify the response of the resonator itself, together with a non-linear model² for the absorption in the absence of flow and a linear vortex shedding model²⁰ to quantify the absorption in the presence of a variety of mean flows.

Theory

Consider the Helmholtz resonator depicted in figure (1). A high amplitude incident sound wave, A , propagates from left to right in a pipe of cross sectional area

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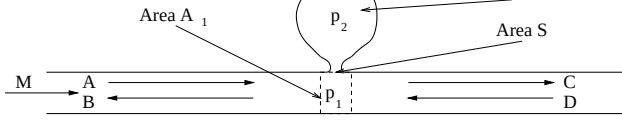


Fig. 1 Helmholtz resonator with incident and transmitted acoustic waves.

A_1 towards a Helmholtz resonator of volume V and neck cross sectional area S where it is partly reflected and partly transmitted into the downstream section of the pipe to give new waves B and C respectively. Downstream of the resonator the transmitted wave C interacts with a downstream boundary to produce a second incident wave $D = RC$, where R is the reflection coefficient describing the downstream boundary to be determined by experiment. A low Mach number flow (which may be zero), of speed U and Mach number M propagates along the pipe. The viscous effects are assumed to be negligible except in the region near the resonator neck.

The stagnation enthalpy is assumed to be continuous at the neck of the resonator:

$$B_1 = A(1+M) + B(1-M) = C(1+M) + R(1-M) \quad (1)$$

Equation (1) is tested experimentally (see Appendix for description of experiment), the results of which are shown in figure (2) where it is seen that the error shown to be small, with the greatest errors found when the kulites or the opening are near a pressure node.

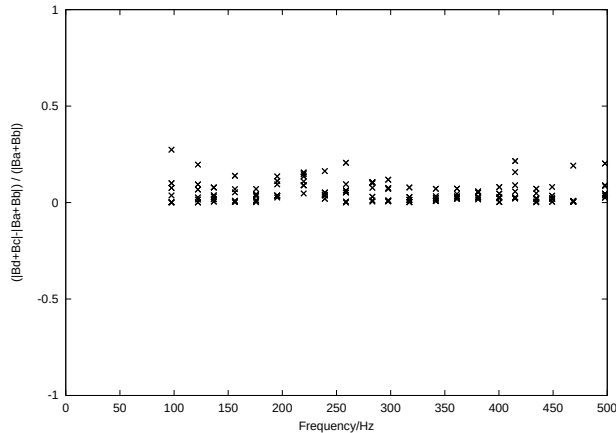


Fig. 2 The change in stagnation enthalpy at the opening non-dimensionalized in terms of the upstream stagnation enthalpy.

The mass flow rate associated with this velocity, $\bar{\rho}Su_1$, is balanced by a change in mass in the resonator chamber:

$$\bar{\rho}Su_1 = V \frac{\partial \rho}{\partial t} = i\omega V \frac{p_2}{\bar{c}^2} \quad (2)$$

where $\bar{c}$ is the mean speed of sound. The variation of $|u_1/p_2|$ with frequency is shown in figure (3), whilst the variation of the phase of u_1/p_2 is shown in figure (4).

and the solid line represents theory. The agreement between the experiment and the theory for the magnitude is good at all frequency with the greatest error occurring near nodes. The agreement for the phase is excellent.

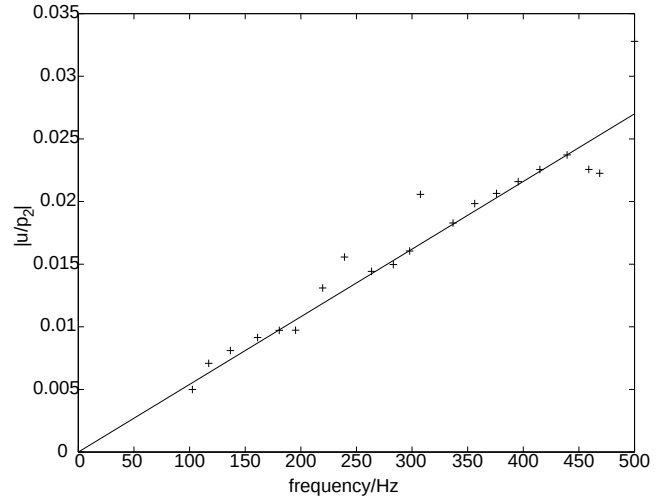


Fig. 3 The variation of $|u_1/p_2|$ with frequency .

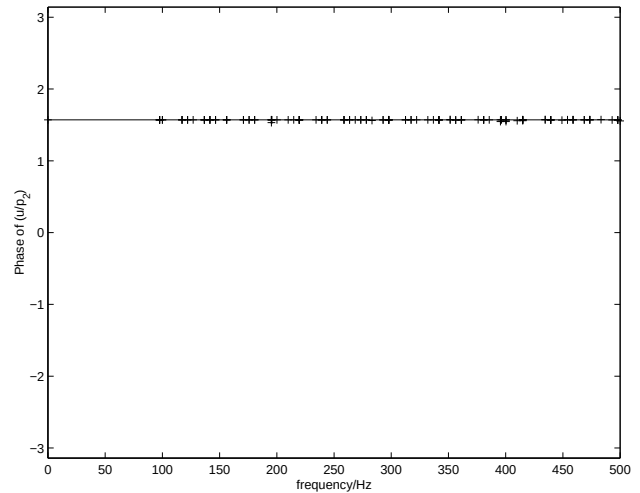


Fig. 4 The variation of the phase of u_1/p_2 with frequency.

The acoustic velocity into the resonator, u_1 , can be related to the acoustic waves in the pipe either side of the resonator using conservation of mass:

$$\bar{\rho}A_1 \left(\frac{A-B}{\bar{\rho}\bar{c}} \right) = \bar{\rho}Su_1 + \bar{\rho}A_1 \left(\frac{C-D}{\bar{\rho}\bar{c}} \right) \quad (3)$$

Equations (1), (2) and (3) relate the acoustic waves to the pressure and velocity in the resonator chamber. In order to determine the response of the system to an incident acoustic wave, we need to relate the acoustic resonator velocity, u_1 , to the pressure difference across it, and hence to the pressure at the opening. This response differs depending upon whether we have a mean flow or not. We consider both of these separately.

In the absence of a mean flow, we use the non-linear model of Cummings^{2,3} and write:

$$p_1 - p_2 = l \frac{\partial u_1}{\partial t} + \frac{c}{2} u_1 |u_1| \quad (4)$$

$$= i\omega l u_1 + \frac{c}{2} u_1 |u_1| \quad (5)$$

for harmonic waves, where ω is the angular frequency, l is the effective length of the Helmholtz resonator neck and c is the discharge coefficient describing the non-linear behaviour of the resonator. Both the effective neck length, l , and the discharge coefficient, c , are to be determined experimentally.

Substituting for p_2 into equation (5):

$$p_1 = u_1 i \left(\omega l - \frac{\bar{c}^2 \bar{\rho} S}{\omega V} \right) + \frac{c}{2} u_1 |u_1| \quad (6)$$

For low amplitude oscillation, the non-linear real part of the right hand side of equation (6) is small and so no absorption of sound energy occurs. The imaginary part then gives resonance at the frequency given by classical theory.

Non-linear equation (6) is solved numerically for the acoustic velocity, u_1 , for a given neck pressure amplitude, p_1 , which can also be related to the downstream waves C and D using equation (1). Equation (3) together with equation (1) can then be solved to give the incident and reflected waves A and B .

The non-linear model developed above is characterized by two parameters, the effective neck length, l , and the discharge constant, c . The former determines the resonant frequency, whilst the latter determines the acoustic absorption. For simplicity we assume that both the effective length and the discharge coefficient are independent of frequency and amplitude of oscillation. As there are two unknowns, the effective length is determined by using a least square fit of experimental data on equation (6), giving an effect length of 30mm; corresponding to overall end corrections of $1.6r$ (based upon the longest portion of the neck). The end correction varies significantly as a result of changes in geometry and flow conditions.^{4,9,26,30,31} Most authors, however, report values between the values of 1.2 and $1.6r$ reported by Rayleigh and Helmholtz. The value obtained here was found to be almost independent of frequency, although some deviation was seen at frequencies below 150Hz. This was probably due to experimental error since the spacing between the kulites is small in comparison with the wavelength at these frequencies and so the processing is more susceptible to measured errors in the phases between kulites. Moreover since frequencies below 150Hz are well below the resonant frequency, errors in the effective length have little effect upon predictions of the absorption. The variation of effective length with amplitude is less than 0.5% except for the highest amplitude studied

was observed resulting in a 5% increase in the resonant frequency. This is consistent with Ih²⁶ who found a slight increase in the resonant frequency with neck pressure amplitude, for relatively long physical neck lengths.

Having determined the effective length of the neck, the linear part of equation (6) can be subtracted from the right hand side to leave the non-linear term. Figure (5) shows the variation of the measured non-linear term with frequency for rig 1. As expected, the non-linear term is only significant near resonance and large neck pressure amplitudes give rise to larger non-linear response. The discharge coefficient can now be determined and is plotted as a function of frequency in figures (6) and (7) for rigs 1 and 2 respectively. Unlike the effective length, the discharge coefficient is not constant and significant variation is seen away from resonance. This variation, however, is almost certainly due to experimental errors since the non-linear term is itself insignificant at those frequencies. Moreover, since the non-linear term is insignificant, any error in the discharge coefficient at these frequencies will not affect predictions of the absorption. To test the sensitivity of the non-linear term to changes in the discharge coefficient we plot the predicted non-linear term as a function of frequency for discharge coefficients of 1.8 and 2.4. The results are shown in figure (8) which is for a sound pressure level at the neck of 135dB on rig 1 with an open-end. Equivalent results for rig 1 with a closed-end and a sound pressure level of 133dB are shown in figure (9). Also shown (solid line) are the experimentally derived values using the effective length obtained from the least square fit. Both values of discharge coefficient give reasonable agreement with experiment, with the experimental falling between the two values. As a result, good predictions should be achieved provided a discharge coefficient of approximately 2 is used.

Figure (7) shows the measured values of the discharge coefficient for rig 2. As higher amplitude neck pressure fluctuations are used, the deviation in discharge coefficient is small for a wider range of frequency and for the different amplitudes. As for rig 1, the measured value is approximately 2. Both these agree well the contraction coefficient, $C_c = \sqrt{\eta/c} \approx \sqrt{1/c}$, for a sharp-edged circular orifice of .66 (see, for example, Massey²⁹). For consistency a constant value of 2 is taken in all the subsequent calculations.

So far the experimental results have confirmed the validity of the assumptions used in the theory and enabled us to specify values for the two characteristic parameters: effective neck length, l , and discharge coefficient, c . Of more interest, however, is the effectiveness of the Helmholtz resonator as an acoustic energy absorber, which we now consider.

Before comparing the predicted and measured ab-

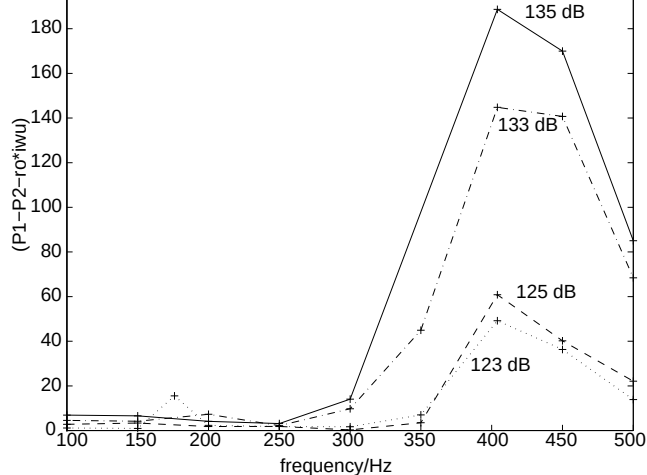


Fig. 5 The difference between the pressure difference and the linear term ($i\omega l u_1$); i.e. the measured non-linear term; as a function of frequency for rig 1.

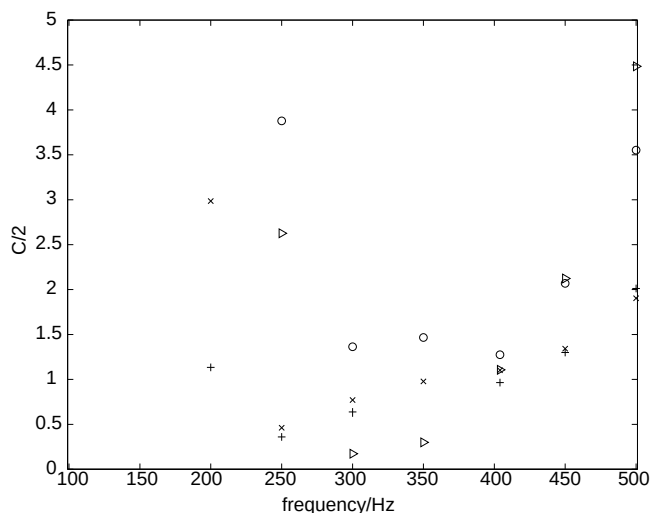


Fig. 6 The predicted value of $c/2$, the non-linear coefficient, as a function of frequency for rig 1 with neck sound pressure levels of 123dB ($\circ$), 125dB ($\triangleright$), 133dB ($\times$) and 135dB ($+$).

sorption coefficients, we first check that the presented theory gives good predictions of the unsteady neck velocity, u_1 . To do this we substitute the values of the effective length, l , and the discharge coefficient, c , obtained above, into equation (6) which is solved iteratively for u_1 . The results are shown in figure (10) which shows results for sound pressure levels of 133dB (rig 1 with a closed-end) and 135dB (rig 1 with an open-end). The figures also show the measured values (dashed line). The agreement is excellent.

Since we now have confidence in the model's ability to predict the unsteady neck velocity, we use equations (1) and (3) together with the unsteady neck velocities we obtain from equation (6) to predict the four acoustic waves; A , B , C and D ; for a range of sound pressure

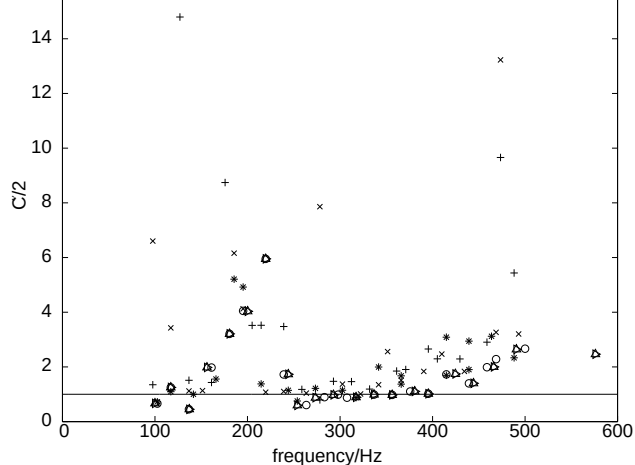


Fig. 7 The predicted value of $c/2$, the non-linear coefficient, as a function of frequency for rig 2 with acoustic neck sound pressure levels of 150dB ($+$), 155dB ($\times$), 160dB ($\circ$), 165dB (Δ) and 170dB ($\triangleright$).

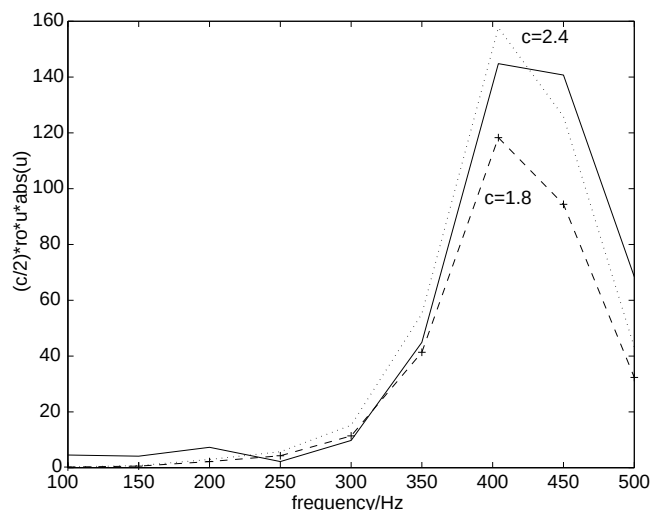


Fig. 8 The effect of varying the value of the non-linear coefficient, C , on the predicted non-linear term for an incident pressure amplitude of 135dB on rig 1 with an open end. The solid line represents the experimental measurements.

levels. As we are interested in the absorption of acoustic energy, we choose to present our results in terms of the absorption coefficient. This is defined as the fraction of incident energy which is absorbed.

$$\text{i.e. } \Delta = 1 - \left(\frac{|B|^2 + |C|^2}{|A|^2 + |D|^2} \right) \quad (7)$$

Figures (11) and (12) show the variation of absorption coefficient, Δ , with frequency for sound pressure levels up to 135dB on rig 1 with an open end and with a closed end respectively. The solid line represents the theory, whilst the dashed line represents the experimental measurements. The agreement is excellent except at low frequencies. There the energy contained in the incident sound field is small. Since

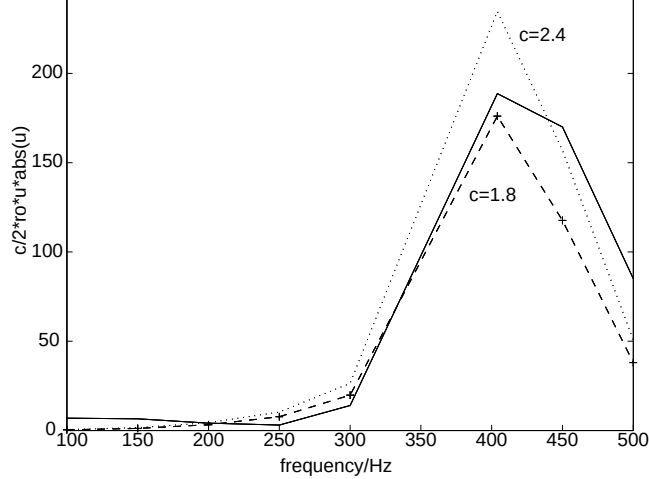


Fig. 9 The effect of varying the value of the non-linear coefficient, C , on the predicted non-linear term for an incident pressure amplitude of 133dB on rig 1 with an closed end. The solid line represents the experimental measurements.

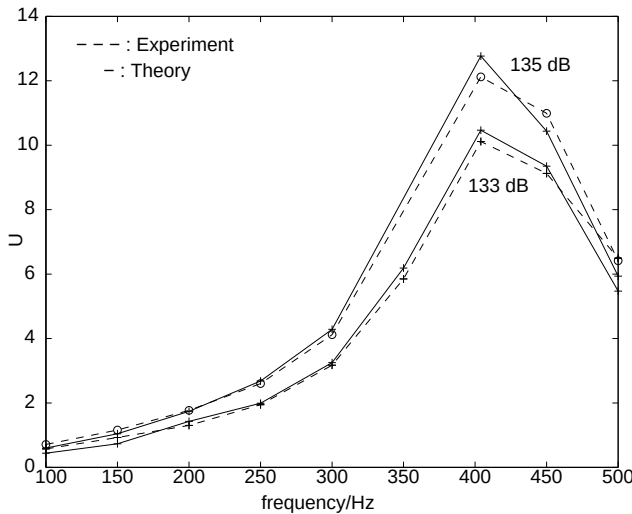


Fig. 10 Comparison between the measured and predicted acoustic velocity, u_1 , as a function of frequency at pressure amplitudes of 133 and 135dB on rig 1.

the absorption coefficient is obtained by dividing by this quantity, a small error in the amount of absorption in the experiment leads to a spuriously large amount of absorption when non-dimensionalized as in figures (11) and (12). This is confirmed by re-plotting the measured absorptions in a dimensional form, i.e. before they are divided by the incident acoustic energy. This is shown in figure (13). Notice that the apparent absorption at low frequencies seen in figures (11) and (12) is absent from this dimensional plot.

Higher amplitude neck pressures were obtained using rig 2 which used a different Helmholtz resonator. Amplitudes of between 155dB and 170dB were measured and comparison between the predicted and measured absorption coefficients is seen in figures (14) —

Figures (11) — (16) suggest that very large absorptions can be obtained; up to 80% for a sound pressure level of 170dB. Even at 133dB the absorption is nearly 50% in a closed-ended pipe of this configuration and nearly 60% for a sound pressure level of 135dB in an open-ended pipe of the configuration used in rig 1. It should be noted, however, that all the results displayed in figures (11) — (16) also include a distinct minimum. This results from a pressure node at the neck of the Helmholtz resonator. This has two effects. The first, and most obvious, is that the sound pressure levels found in practice will be low near this frequency. Secondly, at frequencies near this condition the incident pressure amplitudes need to be large in order to achieve the required sound pressure level. The incident unsteady velocities will, therefore, be very large as they are near an anti-node. The unsteady velocity into the resonator, u_1 , however, is solely a function of the neck pressure amplitude (see equation (6)) and so will not be affected by the presence of the node. Hence the actual amount of acoustic absorption will be small in comparison with the incident sound energy and so the absorption coefficient will be small.

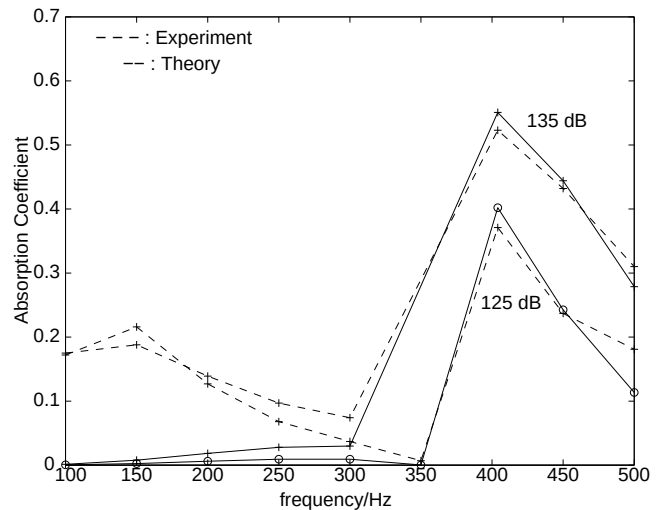


Fig. 11 Comparison between the measured and predicted absorption coefficient as a function of frequency for pressure amplitudes of 125dB and 135dB in rig 1 with an open end.

To confirm the effect of the node, an additional experiment was done on rig 2 with the end left open and with a neck pressure sound pressure of 170dB. As the sound waves are generated in a different way, using unsteady flow through a router, than for rig 1, the physics will change slightly as there will be a small mean flow along the pipe. This is, however, likely to be small, so we expect the behaviour will be similar. Comparison between the predicted and measured absorption in this case are shown in figure (17) where the symbols represent the measured values and the solid line represents the predicted behaviour. As before the

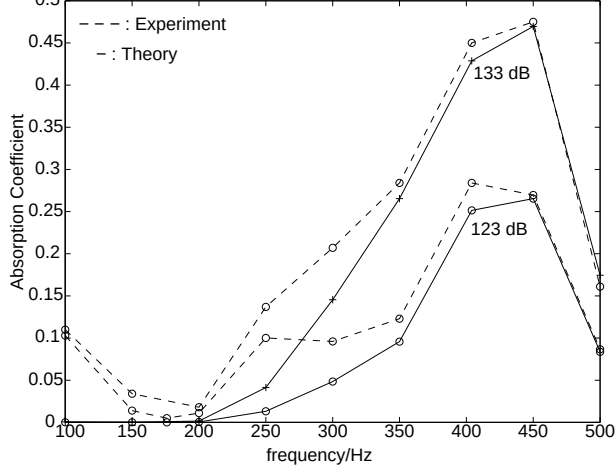


Fig. 12 Comparison between the measured and predicted absorption coefficient as a function of frequency for pressure amplitudes of 123dB and 133dB in rig 1 with a closed end.

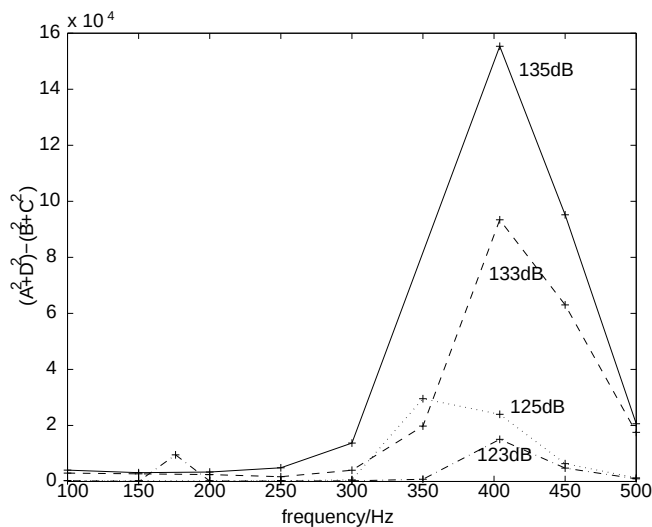


Fig. 13 The variation of the dimensional absorption with frequency for closed and open-ended pipes on rig 1 with pressure amplitudes of 123dB, 125dB, 133dB and 135dB.

agreement is good, despite the small mean flow. The behaviour is also the same with the only main difference being the position of the pressure node which has moved to a frequency of 350Hz.

To summarize the predicted behaviour of the Helmholtz resonator, figure (18) shows the predicted absorption coefficients as a function of frequency for sound pressure levels of between 120 and 185dB, in steps of 5dB. With a sound pressure level at the neck of 120dB, absorption occurs only over a very narrow frequency range and the maximum absorption is less than 10% of the incident acoustic energy. In common with all sound pressure levels, there is a minimum in absorption at a frequency of about 275Hz, which corresponds to the frequency at which a pressure node

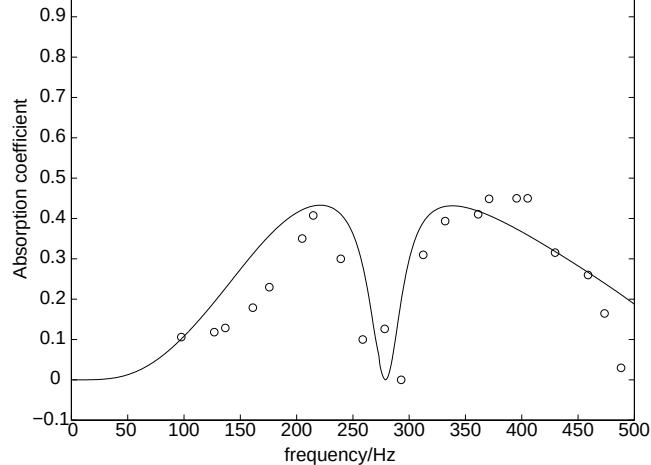


Fig. 14 Comparison between the measured and predicted absorption coefficient as a function of frequency for an incident sound pressure amplitude of 155dB in rig 2 with a closed end.

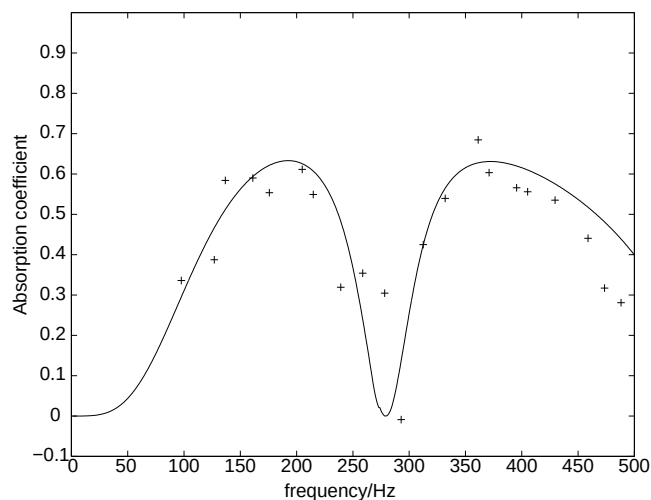


Fig. 15 Comparison between the measured and predicted absorption coefficient as a function of frequency for an incident sound pressure amplitude of 165dB in rig 2 with a closed end.

exists in pipe at the neck opening. This results in no absorption as discussed above. As a result there are two maxima in absorption either side of the node.

As the sound pressure level at the neck is increased, absorption is seen to occur over a broader frequency range, although still with zero absorption at the pressure node. The two frequencies at which peak absorption occurs also move further from the node on either side. In addition the magnitude of the peaks increase. When the sound pressure level reaches 185dB, i.e. just above one atmosphere fluctuation, absorption occurs over a frequency range of about 800Hz, with a peak absorption coefficient of about 82%. As the natural frequency of the Helmholtz resonator is above that of the absorption minimum, notice that the absorption is not symmetric, with the absorption dropping more slowly with frequency as the frequency is increased

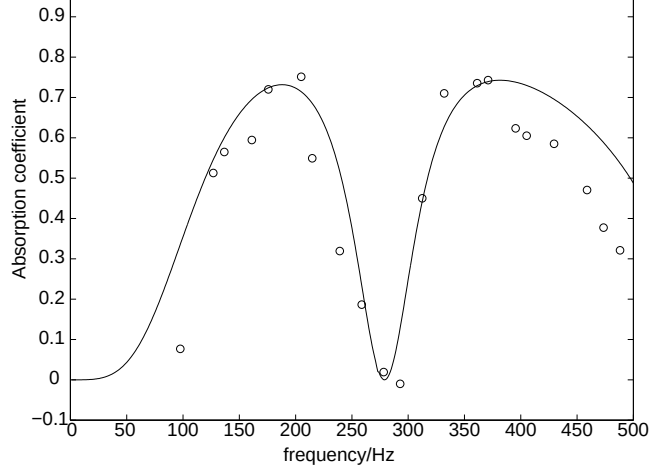


Fig. 16 Comparison between the measured and predicted absorption coefficient as a function of frequency for an incident sound pressure amplitude of 170dB in rig 2 with a closed end.

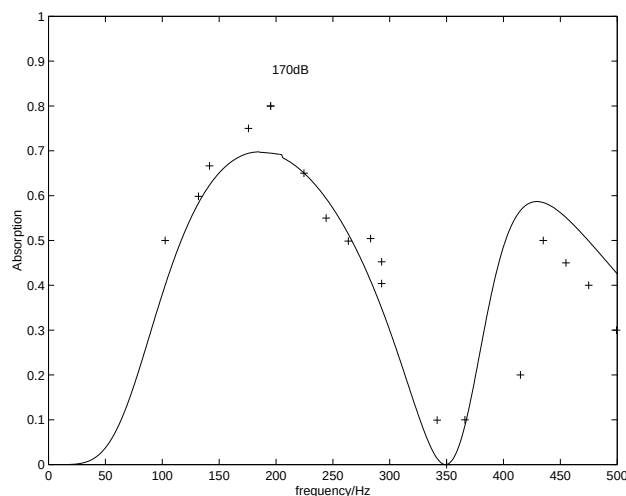


Fig. 17 Comparison between the measured and predicted absorption coefficient as a function of frequency for an incident sound pressure amplitude of 170dB in rig 2 with an open end.

above the second maximum, than is seen by decreasing the frequency below the first maximum. This feature is also evident in the absorption curves for lower sound pressure levels.

In the next section, we discuss the effects of a mean flow.

Linear Theory

In the presence of a mean flow, the acoustical shed vorticity at the trailing edge of the Helmholtz resonator neck opening is swept away by the presence of the flow. Thus a vortex sheet is established which is close to the trailing edge. Such a vortex sheet is a dipole source of sound whose phase relationship with the incident sound energy is such that energy is transferred from the acoustic waves into the vortical part of the flow field where it is eventually dissipated

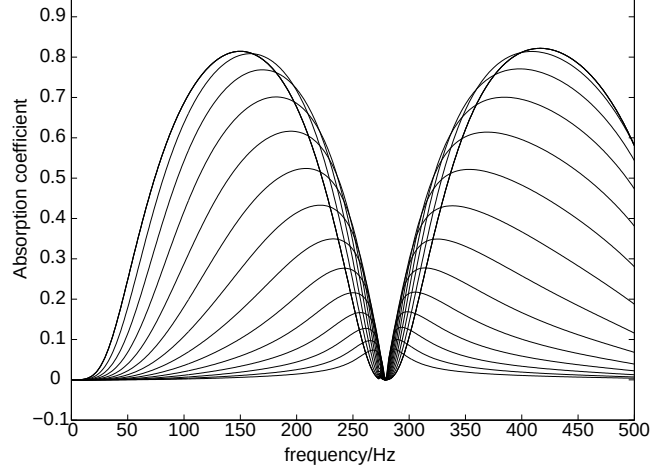


Fig. 18 The predicted absorption coefficient as a function of frequency for a range of incident sound pressure amplitudes from 120dB to 185dB in rig 2.

as heat. This mechanism for acoustic absorption is proportional to the inlet amplitude and so the absorption coefficient is independent of inlet amplitude and can be significant even for modest levels of excitation.^{12-14, 20, 22, 25, 39} Here we consider the response of a Helmholtz resonator to an incident sound field in the presence of three different types of mean flow: cross flow, flow through the neck (bias flow) and a combination of cross flow with bias flow.

Powell³³ identified the Coriolis or vorticity impulse term, $\nabla \cdot (\vec{\omega} \wedge \vec{u})$, as the most significant source of sound in a low Mach number, high Reynolds number flow. Here $\vec{\omega}$ represents the shed vorticity and $\vec{u}$ represents a flow velocity. This is an attractive idea because the source region in which the vorticity is non-zero is a much smaller region than the one in which the Lighthill stress tensor^{27, 28} is non-zero. Moreover, the vorticity field is easier to obtain; such as from CFD, a simple model or experiment. Powell's derivation is not, however, strict in terms of the acoustic analogy.^{27, 28} A more rigorous derivation is given by Howe¹⁸ and Doak⁵⁻⁸ who identify the the stagnation enthalpy, rather than the pressure, as the fundamental acoustic variable.

Howe²⁰ derived a simple model for the vorticity impulse source term in which $\nabla \cdot (\vec{\omega} \wedge \vec{u})$ is linearized and simplified to $\nabla \cdot (\vec{\omega}' \wedge \vec{u})$, where the prime and the overbar represent the fluctuating and mean part respectively. This approximation is valid when the thickness of the unsteadily shed vortex sheet is small in comparison with the thickness of the mean vortex sheet²⁰ and when the mean vorticity strength is small in comparison with the angular frequency.¹² Howe obtains the unsteady vorticity field by assuming that an infinitely thin vortex sheet is shed from the trailing edge whose strength is determined by the Kutta condition. The use of the unsteady Kutta condition in unsteady flows is now well established (see Crighton¹ for

and has been applied to a range of problems including sound absorption by perforated plates,^{20–22, 25, 39} nozzles,¹⁹ linings,^{25, 34} slits^{11, 17, 21, 35} and sudden area changes,^{13, 14} in the presence of either a normal or a tangential mean flow. It is found to agree well with experiments.

Here we use the Powell-Howe concept of vortex sound and Howe's model for the vorticity impulse term for all three flows. The results are quantified, following Howe,^{20, 24} in terms of the Rayleigh conductivity for the neck:

$$u_1 = \left(\frac{p_1 - p_2}{\bar{\rho} i \omega S} \right) \kappa \quad (8)$$

substituting for p_2 we have:

$$u_1 = \left(\frac{p_1 \kappa}{\bar{\rho} i \omega S - i \kappa \frac{c^2 S}{\omega V}} \right) \quad (9)$$

For a known complex function κ , equation (9) can be combined with equations (3) and (1), to obtain the four acoustic waves A , B , C and D . With a mean flow in the pipe, the energy is in terms of the stagnation enthalpy rather than the acoustic pressure³² and so the absorption coefficient is defined as:

$$\Delta = 1 - \frac{|B|^2(1 - M)^2 + |D|^2(1 + M)^2}{|A|^2(1 + M)^2 + |C|^2(1 - M)^2} \quad (10)$$

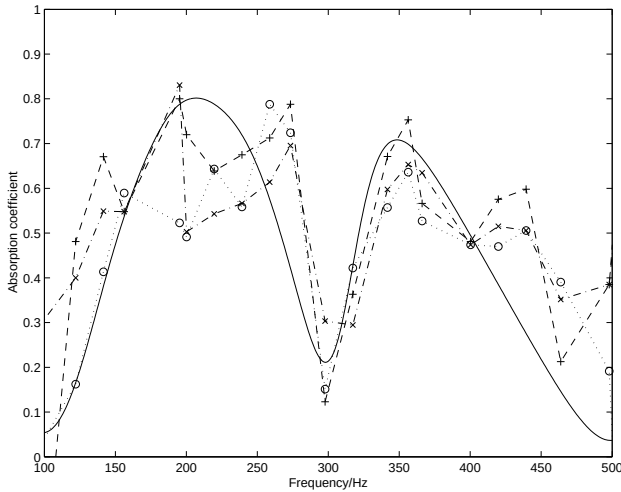


Fig. 19 Comparison between theory and experiment for a uniform mean cross flow of Mach number 0.04; — theory, +- - - experimental measurements at 155dB, x - - - experimental measurements at 150dB, o ···· experimental measurements at 145dB.

Figure (19) shows the measured absorption coefficient as a function of frequency for mean flow Mach number of 0.04 with neck sound pressure levels of 145, 150 and 155dB. The Rayleigh conductivity used in the theory is that calculated by Howe et al²⁴ with a correction to allow for the physical length of the neck. The

pressure amplitude for most of the frequency range at which the absorption is significant. Since the mechanism for sound absorption is now linear, the amplitude of the acoustic velocity in the neck is restricted by a real part of the linear neck impedance. Thus neck sound pressure level needs to be larger to give rise to the large acoustic velocities required for the non-linear mechanism to give significant absorption and so the prediction is good even at these high amplitudes. An absorption coefficient of between 0.4 and 0.6 is achieved for most of the frequency range of interest. Also seen in this plot is reduced absorption near a pressure node, just as we saw in the no flow results.

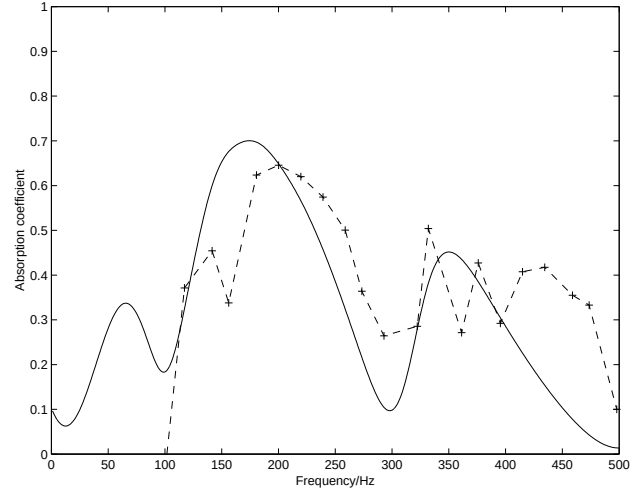


Fig. 20 Comparison between theory and experiment for a uniform mean cross flow of Mach number 0.027; — theory, +- - - (experimental measurements).

Figure (20) shows the measured and predicted absorption coefficient as a function of frequency for mean cross flow Mach number of 0.027. The sound pressure level at the neck is 155dB. Significant absorption occurs over a frequency range of 100 to 500Hz. The prediction is good at most frequencies, although there is a slight discrepancy near the Helmholtz resonator frequency at around 175Hz. This is due to non-linear effects which are only important near the resonant frequency. This is confirmed by figure (21) which shows the variation in the acoustic impedance, u_1/p_1 , with frequency in the presence of a mean cross flow Mach number of 0.027 and at neck sound pressure levels of 135dB to 160dB. For frequencies above about 300Hz, the acoustic neck impedance is almost independent of neck sound pressure level. It is only at frequency around 175Hz that any variation is noticeable. As expected, near this frequency the acoustic impedance reduces with amplitude.

Figure (22) shows the measured absorption coefficient as a function of frequency for mean flow Mach

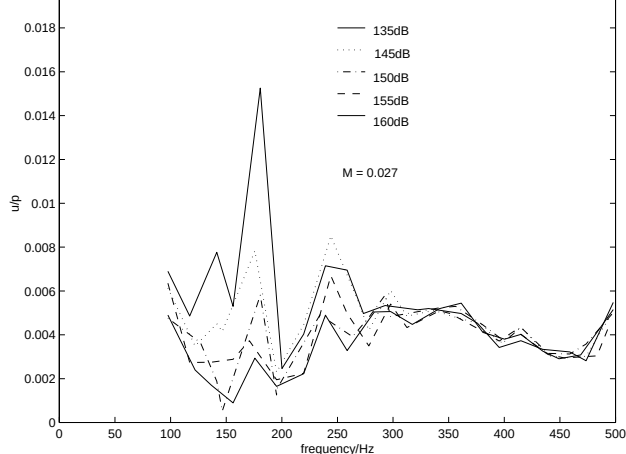


Fig. 21 The measured acoustic impedance, u_1/p_1 , as a function of frequency for neck sound pressure levels of 135dB to 160dB in the presence of a uniform mean flow with Mach number 0.027. — 135dB, 160dB, ... 145dB, ---- 150dB, - - - 155dB

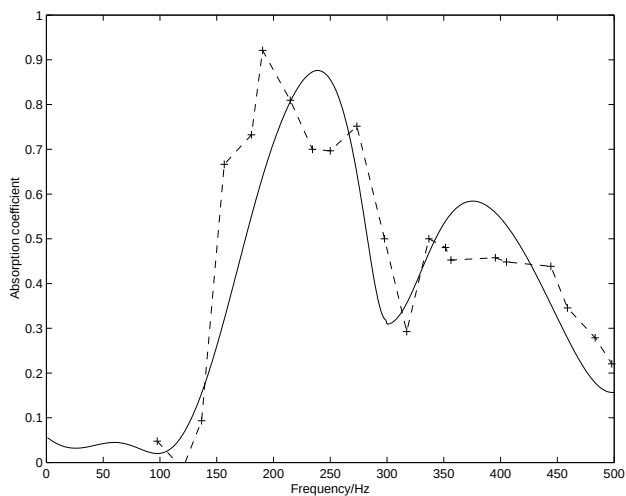


Fig. 22 Comparison between theory and experiment for a uniform mean cross flow of Mach number 0.054; — theory, +- - + (experimental measurements).

number of 0.054. Save for the very high amplitudes, the absorption coefficient is once again independent of amplitude for much of the frequency range. Near resonance, the absorption coefficient is greater than 80% even for a neck sound pressure level of 135dB. As in figures (19) and (20), the presence of a pressure node near 300Hz, is seen to significantly reduce the absorption.

When the flow is a bias flow rather than a cross flow, the results are qualitatively similar as seen in figure (23). Here the Rayleigh conductivity is obtained from Howe's solution for the response of an aperture with a bias flow to acoustical forcing,²⁰ with a correction for the physical length of the neck as for the cross flow effect. There is, however, one crucial dif-

ference with a bias flow. With a cross flow, the trailing edge (i.e. the upstream edge) is associated with the absorption of incident sound due to the coherent shedding of vorticity at the lip. However, when the vortex sheet which convects across the face of the resonator reaches the downstream edge it also interacts with this edge. Since the phase depends upon the distance travelled, the convection speed and the frequency, there will be a range of Strouhal numbers in which this interaction will increase rather than decrease the incident sound field resulting in net generation. This is not observed in figures (19), (20) and (22) for two reasons. First the frequency range for which this occurs is at frequencies higher than shown in the figures. Secondly the physical length of the neck is not insignificant in the experiment and this serves to reduce the generation effect. Nevertheless it is evident simply from blowing across a bottle that Helmholtz resonators can generate sound as well as absorb it. In contrast, when the flow is merely a bias flow, all the vorticity is swept away from the edge and so the resonator only absorbs sound and never generates it.

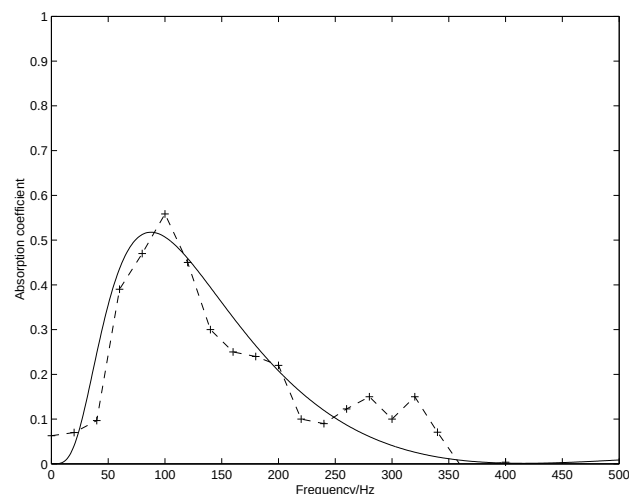


Fig. 23 Comparison between theory and experiment for a uniform mean bias flow of Mach number 0.02; — theory, +- - + (experimental measurements).

A similar effect can be produced by combining a mean cross flow with a small amount of mean bias flow. When this occurs the mean bias flow ensures that the vortex sheet is displaced from the downstream edge when it convects past it and so the interaction is greatly reduced. To quantify this, we perform a new calculation for an aperture in an infinitely thin plate in the presence of a mean cross flow combined with a small mean bias flow. The calculation is based upon Howe et al's calculation for the response with a mean cross flow,²⁴ but with the angle at which the vortex is shed changed to reflect the bias flow and with the imposition of a mean position-dependent displacement.

nus the imaginary part of the Rayleigh conductivity (i.e. the part responsible for production or generation) for cross flow only ($\cdots$), through flow only (---), a bias flow which is 5% of the cross flow velocity (- - -), and 10% through flow (-----). Clearly only a small amount of bias flow is required to destroy the generation of the aperture. Comparison between the measured and predicted absorption coefficient is shown in figure (25), which is qualitatively similar to figure (23).

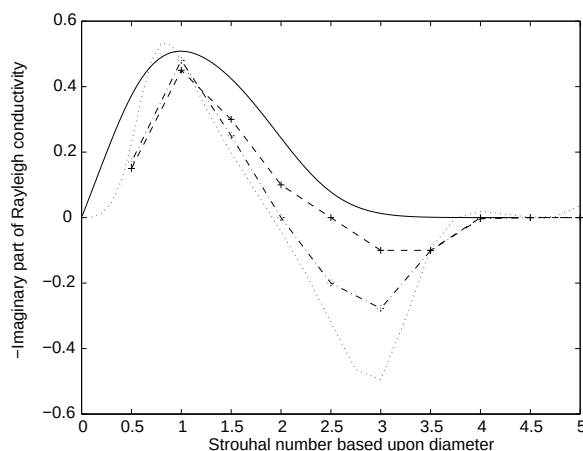


Fig. 24 A graph showing the variation in the imaginary part of the Rayleigh conductivity with Strouhal number for through flow only (---), cross flow only ($\cdots$), 5% through flow (- - -), and 10% through flow (-----).

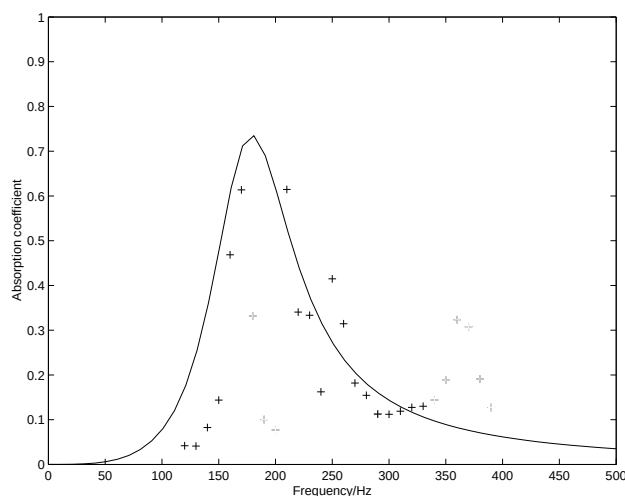


Fig. 25 Comparison between theory and experiment for a uniform mean cross flow of Mach number 0.04 combined with a bias flow Mach number of 0.004; --- theory, + (experimental measurements).

With flow it appears that absorption occurs over a broader frequency range, and that this frequency range is a function of the mean flow Mach number. The absorption is high and independent of acoustic neck pressure amplitude except near resonance where

There is also some evidence that the presence of a pressure node can destroy the absorption just as it did in the non-linear regime.

Conclusions

In this paper we have described in detail the use of Helmholtz resonator as an absorber of acoustic energy. In particular, a theory has been developed, based upon Cummings' non-linear model of an orifice,^{2,3} to describe the acoustical absorption of a Helmholtz resonator at high amplitude with no mean flow. The predictions made using this theory are compared with experiments for sound pressure levels up to 170dB, with good agreement at all amplitudes. The Helmholtz resonator was shown to be an effective absorber of high amplitude sound with absorptions of 75% for sound pressure levels at the neck of the resonator of 170dB. The location of the resonator was, however, shown to be crucial as pressure nodes near the resonator opening result in no acoustic absorption.

In the presence of a mean flow, acoustic absorption occurs over a wider frequency range. The effect of the flow is to sweep the acoustically shed vorticity away from the trailing edge of the neck opening and absorption is achieved which is proportional to the incident acoustic energy (i.e. the absorption coefficient is independent of the amplitude). The linear absorption results in greatly reduced acoustic neck velocities and so the non-linear absorption is not significant except at very high acoustic amplitudes near resonance.

Appendix: Experimental Arrangement

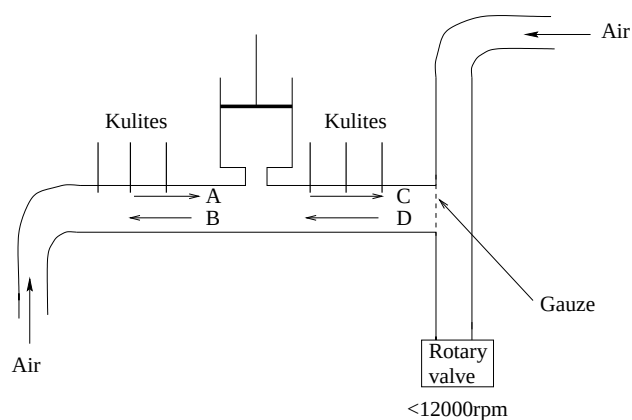


Fig. 26 Experimental rig used for high amplitude incident waves, with an without mean flow.

The experimental arrangement is illustrated in figure (26). Incident acoustic waves are generated in two ways. To produce high amplitudes air is blown through a rotary valve at single frequencies in the range 50Hz to 500Hz (rig 2). At low amplitudes the T-junction and rotary valve is replaced by a loud-speaker (rig 1). The readings from each kulite are processed to give the 4 acoustic waves using the two

viding a check. The kulites are calibrated in the way suggested by Seybert and Ross.³⁶ The spacing of the kulites is chosen to be 11cm which is less than half the wavelength at all frequencies whilst being sufficient to obtain distinguishable measurements. The first kulite is positioned 45mm from the resonator ensuring that the 1st radial harmonic has decayed by at least 62dB by the first kulite.

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